



Impedance characterization of substrate leakage in wafer bonded CMUTs

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Abstract

This work investigates substrate leakage in Capacitive Micromachined Ultrasonic Transducers, (CMUTs) having thin backings. CMUTs produced with solid or rigid substrates have a high Q resonance with flat baseline. However, if the operating wavelength of CMUT becomes larger than the thickness of substrate the real part of cell impedance shows elevated baselines with more leakage at DC than at higher frequencies. When a clamped CMUT membrane is excited harmonically the vibrations produced are counterbalanced by the backing medium. Hence some of the energy is coupled to substrate as loss resulting in a lower mechanical Q of resonance. We first derive this energy coupled to the backing medium using FEA analysis considering the velocity field and associated tensile stresses in a thin backing. Substrate impedance is found numerically from isotropic damping of dispersive elastic waves in the backing. This backing impedance is introduced as a shunt mechanical loss across the transduction force in equivalent circuit model. We characterize the measured impedance of CMUT cells with both FEMs and linear circuit models. Simulations results closely agree with the measured impedance levels, frequency and bandwidth of resonance.

1 Introduction

Mechanical quality factor of a CMUT transducer is determined by the mass of radiating plate and the mass of medium loading it. In airborne applications CMUTs have a higher Q of resonance. The only source of noise effecting the bandwidth of CMUT in air are the internal electrical and mechanical losses. It has been shown analytically in Ünlügedik et al. (2014) that the bandwidth of airborne CMUTs can be increased by employing thin membranes. Thin membranes are easily depressed by atmospheric force over an evacuated cavity resulting in a stiffened membrane. It was shown in Unlugedik et al. (2013) that the plate aspect ratio (radius-to-thickness) must be kept below 35 to avoid geometric nonlinearity in the model. However, this work assumed lossless transduction keeping radiation impedance as the only source of loss in the model.

CMUTs produced through wafer bonding technology in Khan et al. (2020) for beamforming applications at low frequencies and in Yeh et al. (2005) as high frequency linear arrays for medical imaging showed elevated baselines in the real part of measured impedance in air. Similar impedance baselines have been reported before in airborne CMUTs using the same elements for transmission and reception (Ye et al. 2020) and in 3-D volumetric imaging employing a 2-D sparse CMUT array with 128 elements in Choi et al. (2017). This baseline trend effecting the Q of resonance has been reported as leakage to substrate that is higher at low frequencies. It is well known that the passive films have a certain dielectric loss which manifests itself as a baseline at high frequencies in conductance measurements (Khan et al. 2019). However, the dielectric loss does not affect the mechanical Q of plate resonance as much as the substrate leakage and plate frictional loss does. Nonetheless, it is important to consider the contribution of both parasitic capacitance and dielectric losses to overall noise power in reception applications like MEMS microphones.

Substrate loss to backing in form of Rayleigh-Lamb waves was investigated in Badi et al. (2003). The doppler vibrometer measurements revealed presence of both symmetric and antisymmetric lower order Lamb wave modes.

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CMUT devices reported in this paper operated at 2.1-MHz on a 18- μm thick silicon substrate. The devices were characterized by including the resistance of lower order Lamb wave modes into circuit models. Lamb wave modes excited in this work did not produce significant baselines in the measured impedance suggesting that the presence of Lamb waves in solids alone cannot create elevated baselines in the real part of CMUT impedance.

In this paper, we investigate substrate leakage in wafer bonded CMUTs having a 40- μm thick silicon wafer membrane bonded anodically to a 0.5-mm thick borosilicate glass wafer similar to the one used in Khan et al. (2020). The glass wafer consisted of 10- μm deep patterned CMUT cavities. 20 V bias was applied to the bottom electrode through wire bonded connection pads. The membrane is harmonically excited with a 1 V peak amplitude signal applied between electrodes with frequency sweeping from 50 to 100 kHz. Impedance data shows that CMUT plate resonates at 77 kHz. The plate vibrations are counterbalanced by the substrate. This generates longitudinal stresses in the glass medium owing to dispersive elastic waves propagation through the backing. A lower device quality factor with elevated baseline at lower frequencies is indicative of damping loss in glass medium.

To characterize the impedance data, we first evaluate the average power of elastic waves propagating through the backing. The resulting tensile stresses and velocity fields are calculated from FEMs using normal mode theory (Auld 1990). Then considering the transduction force, impedance of elastic waves in glass is derived and introduced in the equivalent circuit model as a shunt mechanical loss (Köymen et al. 2017a). The simulated impedance produced with both circuit and FEM model closely agrees with the measured impedance of CMUT cells.

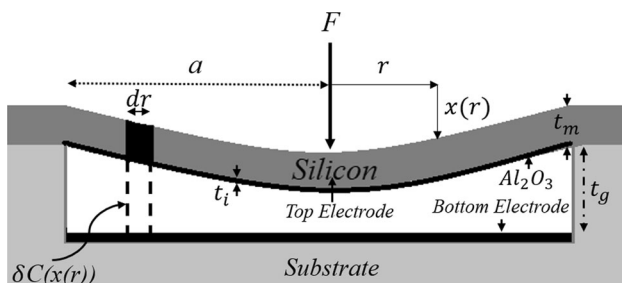


Fig. 1 Cross-section of CMUT cell

2 Linear circuit model for CMUT with substrate loss

A cross-section of wafer bonded CMUT cell is shown in Fig. 1. The membrane is a single crystal silicon having radius a and thickness, t_m . A thin film of alumina of thickness, t_i is present under the membrane to avoid short circuit of cell electrodes. t_g is the etched gap height in substrate medium. The membrane is clamped firmly onto the glass posts but is free to move at the center under the influence of medium loading and applied static. F represents the total force including electrostatic and medium loading required to yield the membrane at its center towards the substrate.

For thin membranes loaded by uniform ambient force the plate profile, $x(r)$ shown in Fig. 1 can be expressed in terms of the radial variable, r as Timoshenko et al. (1959) and Köymen et al. (2012):

$$x(r) = x_p \left(1 - \frac{r^2}{a^2} \right)^2 \quad \text{for } r \leq a \quad (1)$$

where x_p is the peak membrane deflection at its center and $\delta C(r)$ is the capacitance of a small concentric ring having a radius r and width dr given by:

$$\delta C(r) = \frac{\epsilon_0 2\pi r dr}{t_{ge} - x(r)} \quad (2)$$

where $t_{ge} = t_g + t_i/\epsilon_r$ is the electrical length of CMUT gap. This gap height represents the equivalent gap height of an undeflected membrane capacitance ($x(r) = 0$) owing to the series combination of insulation and cavity capacitance. ϵ_0 and ϵ_r are the electric permittivities of gap and insulator material respectively. Note that for thin insulation films having a high relative electric permittivity like alumina or hafnium oxide the electrical length of CMUT gap height becomes equal to the physical gap height, t_g .

To calculate the total capacitance of a CMUT membrane, $\delta C(r)$ is integrated for its full electrode coverage as:

$$C = \int_0^a \delta C(r) dr = C_0 g\left(\frac{x_p}{t_{ge}}\right) \quad (3)$$

where the function $g\left(\frac{x_p}{t_{ge}}\right) = \tanh^{-1}\left(\sqrt{\frac{x_p}{t_{ge}}}\right)/\sqrt{\frac{x_p}{t_{ge}}}$ and the undeflected capacitance $C_0 = \epsilon_0 \pi a^2 / t_{ge}$.

When a time varying harmonic signal $V(t)$ is applied between CMUT electrodes the membrane starts vibrating and generates ultrasound in the medium. Part of this vibrational energy is coupled in the backing as substrate posts tend to counterbalance these vibrations. This creates tensile stress in finite volume meshes of thin substrate medium. The dispersive waves generated in substrate medium have long wavelengths owing to higher

longitudinal and shear speed of sound in solids. The transduction force responsible for introducing the elastic waves in the backing can be written in terms of applied harmonic signal $V(t)$ as Oğuz (2013):

$$f_R(t) = \frac{dE(t)}{dx_R} = \sqrt{5} \frac{C_0 V^2(t)}{2t_{ge}} g' \left(\frac{x_p(t)}{t_{ge}} \right) \quad (4)$$

where $E(t)$ is the instantaneous energy stored in CMUT cell, $E(t) = 1/2C(t)V^2(t)$ and $x_R(t) = x_p(t)/\sqrt{5}$ for the *rms* membrane profile described in (1). The corresponding transduction force, $f_R(t)$, is a source of non-linearity in CMUTs that generates harmonics owing to its squared dependence of the applied harmonic signal $V(t)$. To assess the effect of substrate leakage on receive performance of CMUT cell under small signal conditions the transduction force must be linearized around the static operating point of the membrane as in Koymen et al. (2012):

$$f_R(t) = F_R + f_r(t) \approx f_R|_{x_R=X_R} + \left. \frac{df_R}{dx_R} \right|_{x_R=X_R} x_r(t) \quad (5)$$

where $x_r(t)$ is the *ac* plate displacement. Under small signal conditions, $|x_r(t)| \ll X_R$ implying the *ac* voltage $|V_{ac}(t)| \ll V_{DC}$. That is under linear regime of CMUT operation the transduction force voltage in (4) becomes $V^2(t) \approx V_{DC}^2 + 2V_{DC}V_{ac}(t)$. Expanding (5) and ignoring higher order terms the *ac* part of the transduction force becomes,

$$f_r(t) \approx \frac{2F_R}{V_{DC}} V_{ac}(t) + \sqrt{5} \frac{C_0 V_{DC}^2}{2t_{ge}^2} g'' \left(\frac{X_p}{t_{ge}} \right) x_p(t) \quad (6)$$

where $F_R = \sqrt{5} \frac{C_0 V_{DC}^2}{2t_{ge}^2} g' \left(\frac{X_p}{t_{ge}} \right)$ in (5) is the DC force due to applied bias. $V_{ac}(t)$ is the receive voltage at electrical terminals of CMUT due to incoming ultrasound force, f_{RI} as shown in the receiver circuit model of Fig. 2.

The linear transduction force in model, $f_r(t)$ of (6) is represented by the lumped circuit components including turns ratios, n_R and the spring softening capacitance, C_{RS} . Equation (7) shows how $f_r(t)$ can be expressed in terms of these two lumped circuit components.

$$f_r(t) = n_R V_{ac}(t) + \frac{x_r(t)}{C_{RS}} \quad (7)$$

where

$$n_R = \frac{2F_R}{V_{DC}} \quad (8)$$

and

$$C_{RS} = \frac{2t_{ge}^2}{5C_0 V_{DC}^2 g'' \left(\frac{X_p}{t_{ge}} \right)} \quad (9)$$

The minus sign with spring softening capacitance, C_{RS} indicates opposing direction of electrical transduction force with respect to the membrane restoring force handled by spring compliance capacitance, C_{Rm} . The compliance capacitance, C_{Rm} reacts with the mass of membrane spring, L_{Rm} to produce the mechanical resonance. Z_{RR} is the radiation impedance of a clamped piston radiator (Ozgur-luk et al. 2011). Expressions for C_{Rm} , L_{Rm} and the instantaneous *rms* membrane velocity, $v(t)$ are included in “Appendix”.

2.1 Backing loss in CMUT model

Figure 3 shows a meshed 2D axisymmetric CMUT model constructed in COMSOL Multiphysics (COMSOL Inc., Burlington, MA, USA). Free triangular mesh is used with a minimum mesh size of 0.61 μm and a maximum mesh size of 306 μm (corresponding to 238 elements per wavelength in the backing) to calculate the power coupled in the substrate medium with high accuracy. The CMUT membrane is loaded with a hemispherical air waveguide to represent front radiation medium. Radius of waveguide is set to twice the wavelength of ultrasound in air with its outer surface acting as an absorbing boundary to eliminate any reflections. Both the solid mechanics (solid) and electrostatics (es) physics was used in COMSOL to govern the CMUT cell while the pressure acoustic (acpr) physics was applied to the waveguide and glass substrate. In subdomain of linear elastic materials an isotropic damping factor of 40 was introduced in the glass substrate to dampen the dispersive elastic waves introduced by harmonically excited membrane. Electromechanical Forces (eme) subdomain of Multiphysics was used to model electromechanical interaction between CMUT structure and electric fields while Acoustic-Structure Boundary (asb) coupling was used to couple the Pressure Acoustics domain with solid CMUT

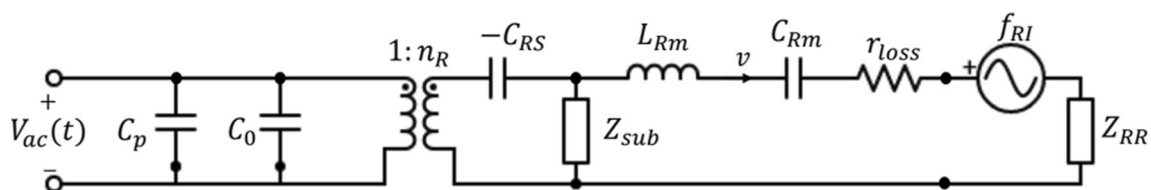


Fig. 2 Small signal model including the shunt backing loss, Z_{Sub}

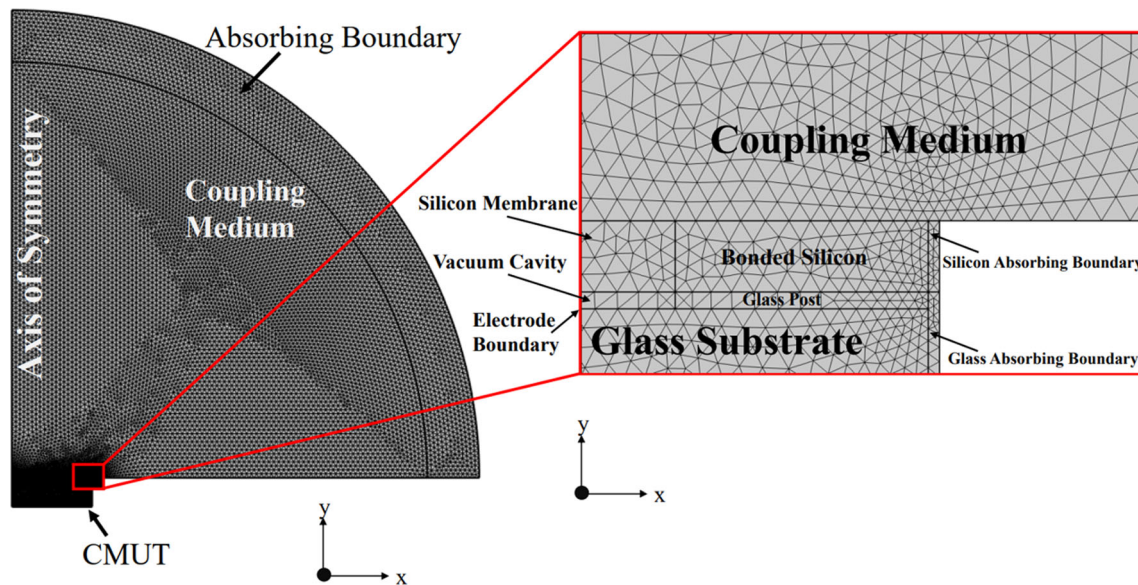


Fig. 3 2D axis-symmetric meshed CMUT model in COMSOL multiphysics

structure. Material and dimensional parameters of CMUT are given in Table 1.

To derive the mechanical loss associated with damping of elastic waves in the backing, we first determine the average normal power per unit length, P_l coupled by the vibrating membrane to the substrate in terms of tensile stresses, $T(x, y)$ and normal velocity of elastic waves; $u(x, y)$ as in Badi et al. (2003) and Auld (1990):

$$P_l = -\frac{1}{2} \int_{-l}^0 \left(u_x^* T_x + u_y^* T_{xy} \right) dy \quad (10)$$

where the normal velocity and tensile stress parameters u_x , u_y , T_x and T_{xy} are obtained from FEA at the glass post boundary (where the silicon wafer is bonded in Fig 3). The average power of the normal wave mode propagating through the CMUT backing becomes:

$$P_{av} = |a|^2 P_l \quad (11)$$

The coefficient a represents the normal mode amplitudes of velocity fields and stress distributions which is expressed as:

$$a = -\frac{K}{4P_l} \int_{-l}^0 \left(u_x^* T_x + u_y^* T_{xy} + u_x T_x^* + u_y T_{xy}^* \right) dy \quad (12)$$

and K depends on the isotropic damping factor of glass medium. A lossy glass section at the end of the substrate creates a perfect matched boundary by eliminating any reflections at the lossy glass interface (Yaralioglu et al. 2000; Moulin et al. 2000). To be consistent with the circuit convention of Fig. 2, the substrate loss, $Z_{sub} = R_{sub} + jX_{sub}$ appearing across the transduction force, f_R in the model is expressed as:

$$Z_{sub} = \frac{f_R^2}{P_{av}} \quad (13)$$

We obtain the substrate impedance, Z_{sub} from (13) using the transduction force, f_R required to propagate the normal elastic wave modes into the backing medium with power P_{av} . Figure 4 shows the power, P_{av} coupled by the vibrating membrane into the substrate for a CMUT geometry listed in Table 1. Real part of the power, P_{av} coupled to the substrate increases with frequency resulting in a substrate loss that is higher at low frequencies and then decreases at higher frequencies. The substrate loss, Z_{sub} is shown in Fig. 5. Z_{sub} appear across the transduction force as a shunt mechanical loss in the circuit model. The real part of substrate impedance, R_{sub} is high at lower frequencies but decreases with frequency. This damping loss

Table 1 Geometrical parameters of CMUT

a	Radius	1.4 mm
t_m	Membrane thickness	40 μm
t_{ge}	Effective gap height	10 μm
t_i	Insulation layer	100 nm
V_{DC}	DC bias	20 V
F_b/F_g	Normalized static force	0.75
Y_0	Young's Modulus	148 GPa
ρ_m	Si plate density	2370 kg/m ³
σ	Poisson's ratio	0.17
ϵ_r	Dielectric constant	9.5

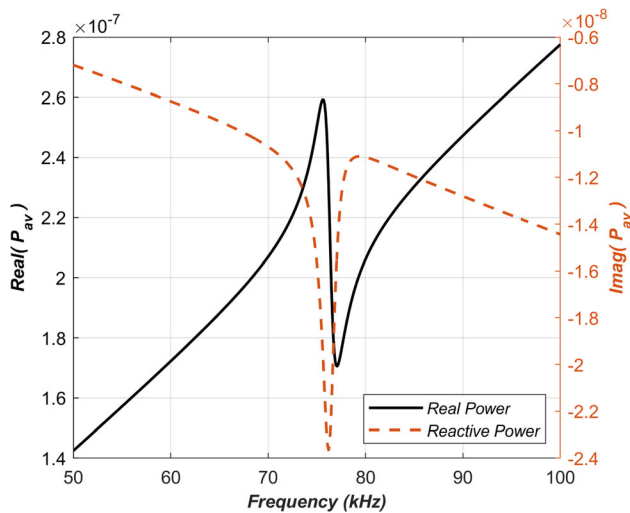


Fig. 4 Power coupled into the substrate by a harmonically excited CMUT membrane

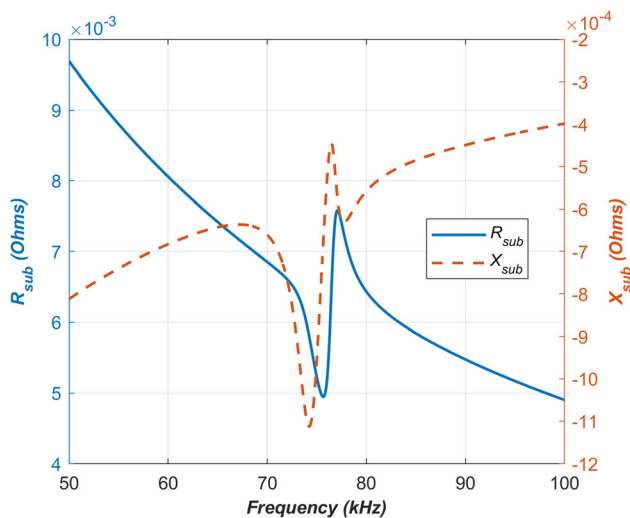


Fig. 5 Substrate impedance, Z_{sub}

of waves in glass is responsible for creating elevated baselines in the measured real part of CMUT impedance. X_{sub} represents the reactive part of thin glass backing.

2.2 Open circuit receive voltage in CMUTs with thin backing

The open circuit receive voltage response to the incoming dynamic force, f_{RI} in the model can be compared to CMUTs with thick or rigid substrates. CMUTs with thick substrates have a large shunt impedance equivalent to an open circuit across the transduction force in the circuit model of Fig. 2. Circuit model of Fig. 2 is simulated in Advance Design Systems (ADS, Keysight Technologies, Santa Rosa, CA.) Magnitude of the open circuit receive

voltage for CMUTs having thin backings is compared with a rigid backing in Fig. 6.

CMUTs having thin backings have a lower open circuit voltage owing to substrate leakage. This leakage is more dominant when the membrane is biased close to the substrate as can be seen for a 200 V biasing case. The difference in the open circuit receive voltage is more for large biasing. The spring softening effect is not observed in CMUTs with leakage to substrates. Figure 7a shows this phenomenon captured in the measured resistance of two CMUT cells having the same dimensional parameters listed in Table 1. As can be seen the resonance frequency does not shift with the bias. The leakage to substrate is evidenced in the elevated impedance baselines at lower frequencies.

3 Fabrication of wafer bonded CMUTs

For impedance characterization of wafer bonded CMUT cells with thin glass backings we employ the same CMUTs used in Khan et al. (2020) as shown in Fig. 8. A 420 μm thick SOI wafer was first passivated by a 100 nm thick film of Aluminium oxide deposited on the device side using Atomic Layer Deposition (ALD) process. SOI wafer was diced before the bonding to expose CMUT connection pads (Fig. 8b). The diced silicon wafer was anodically bonded with a 500 μm thick glass wafer consisting of 10 μm deep circular cavities. The glass wafer was patterned using a chrome feature mask in the lithography process. A buffered oxide solution, BOE (1:7) was used to etch circular cavities in the glass followed by metallization of etched features in an e-beam evaporation chamber. The bottom CMUT electrodes consisted of a 250 nm thick metal stack of Ti

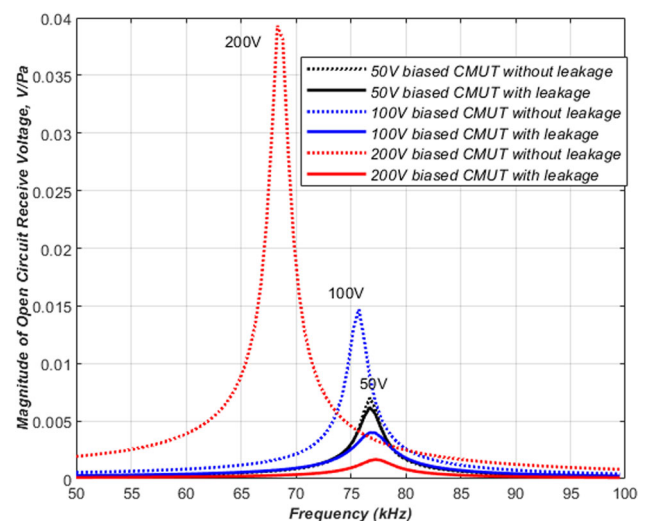


Fig. 6 Comparison of open circuit receive voltage response in CMUTs with thin backings

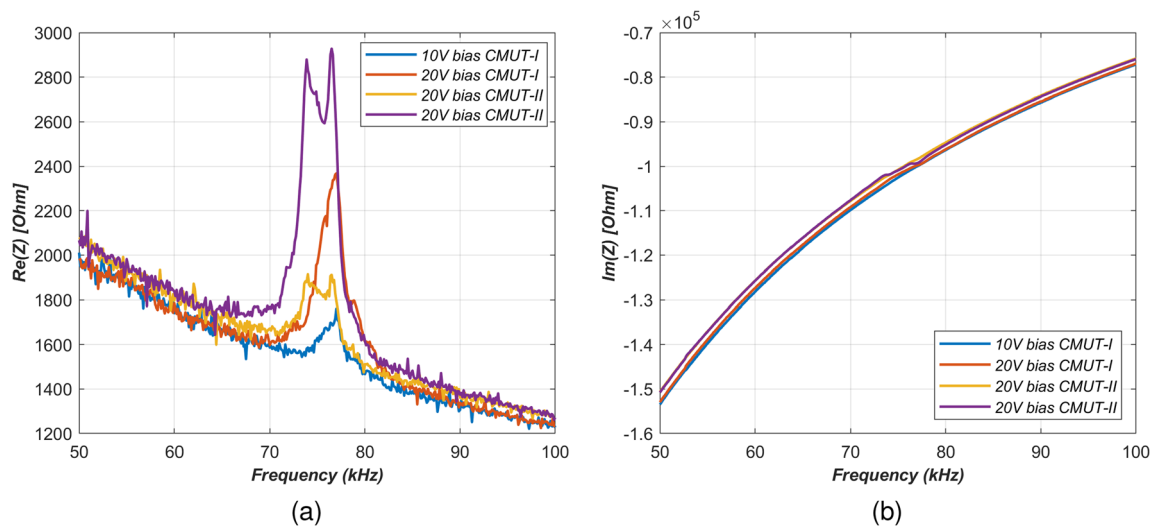


Fig. 7 Measured impedance of two identical CMUT cells (CMUT-I and CMUT-II) on a 0.5 mm glass substrate, **a** real part and **b** imaginary part of impedance

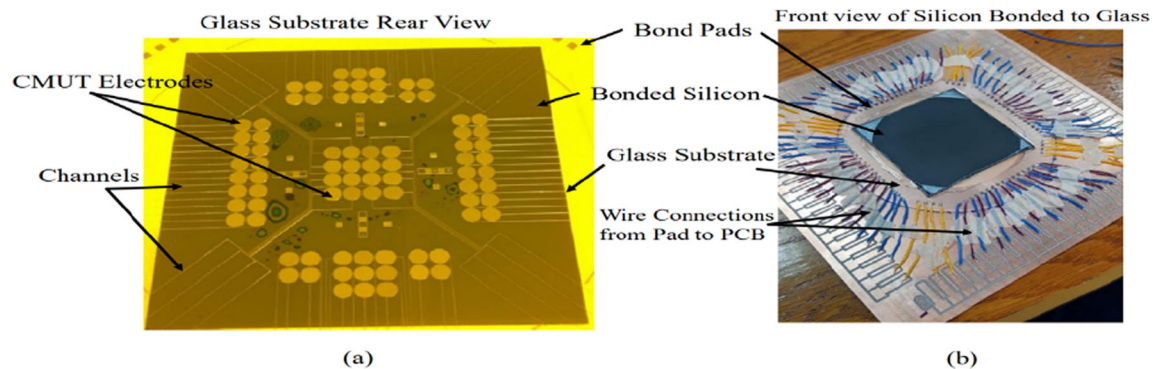


Fig. 8 Wafer bonded CMUTs, **a** glass side view, **b** wire bonded CMUT cells

and Au. Lift-off process in piranha solution was followed to remove the residual resist and metal off the glass wafer. The diced and passivized SOI wafer was then anodically bonded with glass wafer. The handle layer of SOI wafer was dry etched using Reactive Ion Etching (RIE) process. The dry etching of silicon stopped at the buried oxide layer of bonded SOI wafer which was later removed using BOE(1:7) wet etchant.

CMUT channels were finally sealed by applying a sealing epoxy and venting out trapped air from CMUT cavities in a vacuum chamber followed by epoxy curing.

4 Experimental characterization of substrate loss

The input impedance of CMUT cells was measured using a HP 4194A impedance analyzer. Figure 9 shows the real part of measured and simulated impedance of two identical CMUT cells having geometrical parameters listed in

Table 1, (CMUT-I and CMUT-II). Impedance analyzer cables were first calibrated out to measure pure CMUT impedance. Both cells were excited with a 1 V peak amplitude harmonic signal with frequency sweeping from 50 to 100 kHz and an applied bias of 20 V. Channel parasitic capacitance, C_p of 13 pF was added across the CMUT capacitance, C_0 to match the measured reactance of CMUT cells. To match the bandwidth of resonance at 77 kHz a series loss, $r_{loss} = 2.56S\rho c$ for CMUT-I and $r_{loss} = 1.58S\rho c$ for CMUT-II was added in series with radiation impedance in the circuit model to account for plate frictional loss due to residual air left in cavities. Where $S = \pi a^2$ is the area of CMUT radiation plate and ρc is the plane wave impedance in air.

Alumina dielectric loss was modelled by a shunt conductance with a loss tangent of 0.00014 across the CMUT capacitance, C_0 . We observed that the dielectric loss does not affects the measured mechanical quality factor much or the baseline in the real part of impedance data above 50 kHz. The backing impedance, Z_{sub} shown in Fig. 5 was

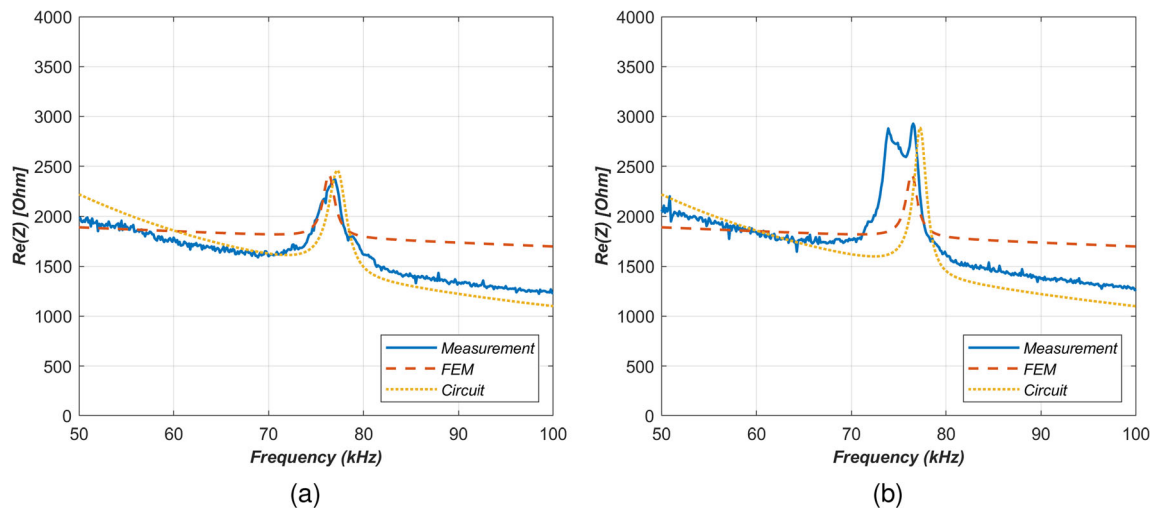


Fig. 9 Measured and simulated impedance of **a** CMUT-I and **b** CMUT-II

added across the transduction force as a shunt mechanical loss. The real part of Z_{sub} produces elevated impedance baseline in the circuit model that characterizes the measured impedance shown in Fig. 9.

5 Results and discussion

The simulated bandwidth of resonance and frequency closely matches with the measured impedance of CMUT-I. For CMUT-II the simulated resonance is 2% higher, this can be attributed to a higher spring softening effect due to applied bias owing to a deeper etched cavity in CMUT II or variation in plate thickness or material properties from those listed in Table 1. Simulated bandwidth of CMUT-II is 2 kHz lower in FEMs. This is attributed to asymmetrical shape of CMUT-II resonance which cannot be excited by uniform excitation if the CMUT had completely symmetric geometry. However, due to production inaccuracies, a symmetrical excitation can still impair the shape of resonance.

In both FEM and circuit models any reflections from nearby cavities in form of surface waves is ignored. Since multiple CMUTs are produced in closely packed arrays with varying distances the effect of mutual cross talk through substrate is not considered in this work. Impedance of surface waves and substrate cross talk might contribute to the difference in simulated baselines, but an accurate assessment of these effects is difficult owing to large number of cavities and the location of CMUT cell and its distance from other cells. Based on the circuit and FEM simulations the elevated baselines that tend to decrease with frequency is reasonably reproduced showing strong presence of elastic waves in the backing medium. This implies CMUTs not only radiate ultrasound in the front

surrounding medium but also couples energy into the backing with a different wave impedance.

Wafer bonded CMUTs have clamped membrane conditions at the rim of radial substrate post where the silicon plate is bonded. Hence the static plate displacement is zero at the rims, but the AC particle velocity is not zero. This boundary condition introduces elastic waves into the substrate medium in form of tensile stresses. These elastic waves suffer less damping at low frequencies resulting in more leakage to substrate at DC. Substrate loss decreases with frequency but also lowers the device quality factor. The real part of substrate loss R_{sub} shown in Fig. 5 reproduces the measured impedance baseline in the model. This wave impedance should be included as a mechanical loss in the circuit model to simulate correct bandwidth and impedance of CMUTs produced on thin substrates.

6 Conclusion

In this work the backing loss in CMUTs having thin substrates is investigated. The real part of measured CMUT impedance shows more leakage at lower frequencies. This loss effects the quality factor of resonance and lowers the open circuit receive voltage of CMUT as a receiver. This loss is characterized by considering the transduction force and the average power of elastic waves propagating into the backing. The radiation impedance of elastic waves in backing should be introduced as a shunt impedance branch across the transduction force as a damping loss in equivalent circuit model to reproduce the measured bandwidth of resonance and elevated baselines in the real part of impedance data.

Appendix: Mechanical lumped circuit components

In the small signal model of Fig. 2 the lumped inductance, L_{Rm} represents the mass of the membrane having density ρ :

$$L_{Rm} = \pi a^2 \rho t_m \quad (14)$$

and C_{Rm} is the membrane compliance expressed in terms of its Young's modulus, Y_0 and Poisson's ratio, σ as:

$$C_{Rm} = \frac{9(1 - \sigma^2)a^2}{80\pi Y_0 t_m^3} \quad (15)$$

The fundamental mode of membrane resonance can be expressed in terms of mechanical lumped circuit components L_{Rm} and C_{Rm} as:

$$f_m = \frac{1}{2\pi\sqrt{L_{Rm}C_{Rm}}} = \frac{2}{3\pi} \frac{t_m}{a^2} \sqrt{\frac{5Y_0}{(1 - \sigma^2)\rho}} \quad (16)$$

The *rms* current through the mechanical section of linear circuit is,

$$v(t) = \frac{dx_R(t)}{dt} \quad (17)$$

In the absence of radiation reactance and substrate leakage the series resonance of *rms* circuit model is given by Köymen et al. (2017b):

$$f_s = \frac{1}{2\pi\sqrt{L_{Rm}\left(\frac{1}{C_{Rm}} + \frac{1}{-C_{RS}}\right)^{-1}}} \quad (18)$$

$$= f_m \sqrt{1 - \frac{2}{3} \left(\frac{V_{DC}}{V_r}\right)^2 g'' \left(\frac{X_p}{t_{ge}}\right)}$$

where V_r is the vacuum collapse voltage:

$$V_r = 8t_{ge} \left(\frac{t_m}{a}\right)^2 \sqrt{\frac{Y_0 t_{ge}}{27\epsilon_0 t_m (1 - \sigma^2)}} \quad (19)$$

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Author contributions TMK made the impedance measurements at BASTA Lab and MK characterized the substrate leakage using FEA analysis and equivalent circuit models.

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Declarations

Conflict of interest The authors declare no conflicts of interest of any kind.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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